

Week 1 - Day 1b: Strings and Booleans

Welcome to the exercises for Week 1, Day 1b! Today, you'll practice the concepts we covered in lecture 1b. We talked about strings, booleans, and types, and we also briefly introduced variables. If you find variables confusing, don't worry — we'll explore them in more detail tomorrow.

Useful Tips:

- Jupyter consists of a series of cells
- Use shift + enter to run a cell
- Remember the order of operations!
- Remember that types can change after an operation!

Exercise 1: Strings `[]`

In this exercise, we will practice creating and working with strings. A *string* is a sequence of characters, which can include letters, numbers, punctuation marks, and spaces.

Let's start by defining some strings below and exploring their properties.

```
# Creating strings
print("Hello AddisCoder")    # A simple greeting
print("Saba")                # A name
print("This is a string.")   # A sentence
print("1234")                # A string of numbers
print("!@#$$%^&*()")        # A string with special characters
```

Now it's your turn. We'll practice creating strings below.

1.1 Create the string `Hello`

```
# TODO: Your code here
```

1.2: Create the string `AddisCoder!`

```
# TODO: Your code here
```

Exercise 2: Basic String Operations `[]`

In this exercise, we will learn how to perform basic operations on strings. One of the most common operations is **string concatenation**, which just means joining two or more strings together.

Hint: Use the `+` operator and add two strings to each other as if you are adding two numbers.

2.1 Concatenate the strings `"Hello"` and `"AddisCoder!"`

```
# TODO: Your code here
```

2.2 Concatenate the strings "Hello" and " AddisCoder!"

```
# TODO: Your code here
```

2.3 Concatenate the strings "Hello " and "AddisCoder!"

```
# TODO: Your code here
```

Exercise 3: Exploring String Contents []

You can put all sorts of things within strings. This can be very confusing! Make sure you know what is inside a string vs not.

3.1 Concatenate the strings "Hello + " and "Addis Coder!"

```
# TODO: Your code here
```

3.2 Concatenate the strings "Confusing ^ * " and "string"

```
# TODO: Your code here
```

3.3 What else can you put in strings? Define a few of your own below.

```
# TODO: Your code here
```

Exercise 4: Converting Numbers to Strings []

We can convert numbers to strings. Do you remember the syntax?

Hint: Use the `str` function. For example, `str(5)` converts the variable `5` of type `int` to variable `"5"` of type `string`.

Important Note: `str` is an *programming function* as opposed to a *mathematical functions*. It takes an input of any type and returns an output of type `string` for the corresponding variable. Earlier today, we saw two other *programming functions*: `type` and `print`. (Recall `type(3.14)` from Exercise 5 from 1a.) `type` returns the data type of its input and `print` is a function that commands the terminal to print out its input. Keep this in mind for Week 1 Day 3!

4.1 Convert `3` into a string

```
# TODO: Your code here
```

4.2 Convert `3.1415` into a string

```
# TODO: Your code here
```

4.3 Convert 27891 into a string

```
# TODO: Your code here
```

Exercise 5: Combining Strings and Numbers []

Strings and numbers are not the same *type*. However, they can be used together in certain operations, depending on the operation and ordering. This can be very confusing!

For each of the following, predict what the output will be. Then, run the code in Jupyter to check your answer.

5.1 What will happen when you concatenate a string with a number represented as a string? What will the output look like, and what would be its type? Here is some data types for your reference:

```
type(1)          == int
type(10000)       == int
type(3.14)        == float
type(2.0)         == float
```

```
# TODO: Put your guess here before running the code cell
```

Run the following cell to check your answer.

```
"hello" + "3"
```

Useful Tip: You could also check its type by running the following code. You can use this strategy for all the questions below if you want!

```
type("hello" + "3")
```

5.2 What will happen when you try to concatenate a string and an integer? What will be the output look like *if any*, and what would be its type *if any*?

```
# TODO: Put your guess here before running the code cell
```

```
"hello " + 3
```

5.3 What will happen when you concatenate a string with the string representation of a number?

```
# TODO: Put your guess here before running the code cell
```

```
"hello" + str(3)
```

5.4 What will happen when you multiply a string by an integer?

```
# TODO: Put your guess here before running the code cell  
"hello" * 3
```

5.5 What about this?

```
# TODO: Put your guess here before running the code cell  
"hello " * 3
```

5.6 What will happen when you multiply an integer by a string?

```
# TODO: Put your guess here before running the code cell  
3 * "hello"
```

Exercise 6: Booleans []

We also covered *booleans*. Recall that Booleans represent one of two values: `True` or `False`. Compute the following using Jupyter to check the **type of the output**.

6.1 `type(True)`

```
# TODO: Your code here
```

6.2 `type(False)`

```
# TODO: Your code here
```

6.3 `type("True")`

```
# TODO: Your code here
```

6.4 `type("False")`

```
# TODO: Your code here
```

Exercise 7: Boolean Case Sensitivity []

It's very important that `True` and `False` are capitalized correctly.

What happens if you run the following in Jupyter:

7.1 `true`

```
# TODO: Your code here
```

7.2 `false`

```
# TODO: Your code here
```

7.3 True

```
# TODO: Your code here
```

7.4 False

```
# TODO: Your code here
```

Exercise 8: Equality Comparison []

We can compare items and the result of a comparison is a boolean. The first comparison operation we learned was `==`

Guess what the output will be for the following and check with Jupyter:

8.1 `1 == 1`

```
# TODO: Put your guess here before running the code cell
```

Now check your answer!

```
# TODO: Check your guess here
```

8.2 `1 == 2`

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

8.3 `True == True`

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

8.4 `True == False`

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

8.5 `"hi" == "hi"`

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

```
8.6 "hi" == "bye"
```

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

```
8.7 "hi" == "Hi"
```

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

Exercise 9: Comparing Different Types []

There are some confusing ways to compare things in Python.

Guess what the output will be for the following and check with Jupyter:

```
9.1 1 == True
```

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

```
9.2 True == 1
```

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

```
9.3 1 == "1"
```

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

```
9.4 "1" == True
```

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

Exercise 10: Inequality and Comparison []

We can also compare for inequality, less than and greater than.

Guess what the output will be for the following and check with Jupyter:

10.1 $1 < 1$

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

10.2 $1 > 1$

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

10.3 $1 < 2$

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

10.4 $1 \leq 1$

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

10.5 $1 \leq 2$

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

10.6 $2 \geq 2$

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

10.7 $1 \neq 1$

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

10.8 $1 \neq 2$ (Hint: $!$ is like a negation, meaning **not**.)

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

Exercise 11: Comparing Strings and Other Types []

In this exercise, we will explore how comparisons work not only with numbers but also with other types such as strings and booleans. Comparisons with strings and booleans can sometimes produce unexpected results, so let's see how Python handles these cases.

Guess what the output will be for the following comparisons and then check your answers by running the code in Jupyter:

11.1 `"hello" < "bye"`

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

11.2 `"hello" > "bye"`

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

11.3 `"Hello" > "bye"`

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

11.4 `False < True`

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

11.5 `3 != "3"`

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

11.6 `3 <= "3"`

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

Exercise 12: Boolean Logical Operators with AND []

There are also special operations between booleans. The first one we learned was `and`.

Guess what the output will be for the following and check with Jupyter:

12.1 True and True

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

12.2 True and False

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

12.3 False and True

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

12.4 False and False

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

Exercise 13: Boolean Logical Operations with OR []

The second operation between booleans is the **or** operation.

Guess what the output will be for the following and check with Jupyter:

13.1 True or True

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

13.2 True or False

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

13.3 False or True

```
# TODO: Put your guess here before running the code cell
```

```
# TODO: Check your guess here
```

13.4 False or False

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

Exercise 14: Type Conversions in Operations []

One very confusing thing about Python is how different types are converted to each other.

Guess what will happen with the following and check your answer with Jupyter:

14.1 3 + True

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

14.2 True * 3

```
# TODO: Put your guess here before running the code cell  
# TODO: Check your guess here
```

Congratulations []